Hw2_part1_CV Verification System Report

Name: Liao Ruiqi
Student ID: 1155245575

1. System Architecture and Design Decisions

The automated CV verification system is designed as a modular, asynchronous pipeline powered by Large Language Models (LLMs) and the Model Context Protocol (MCP). The architecture prioritizes fault tolerance, structured data processing, and deterministic scoring.

Core Architectural Components

- LLM Integration: The system utilizes Google's Generative AI (gemini-2.0-flash) as the core reasoning engine. It relies heavily on prompt engineering to force structured JSON outputs, converting qualitative analysis into programmatic variables.
- MCP Client Adapter: The system interfaces securely with the provided SocialGraph MCP server via MultiServerMCPClient. This decouples the agent's logic from the underlying API implementations of the social media platforms.
- Asynchronous Pipeline (asyncio): To optimize processing time across 5 CVs while respecting API rate limits, the system leverages asynchronous function calls (ainvoke).

Key Design Decisions

- Structured Data Funnels: Instead of processing raw text end-to-end, the system enforces a strict data funnel. Unstructured PDF text is immediately converted into a structured JSON schema (Name, Location, Education, Experience, Skills) before any tool is called.
- LLM as a "Jury" for Profile Matching: Rather than relying on rigid string-matching algorithms for names or locations, the system passes the top 3 candidate profiles retrieved by MCP tools to the LLM. The LLM acts as a jury, evaluating semantic similarity (e.g., nicknames, synonymous job titles) to select the true candidate profile.
- Robust Fallback Parsing: Because LLMs occasionally wrap JSON outputs in Markdown blocks (e.g., ```json), a custom parse_llm_json_response function was implemented using standard Python parsing and regex fallbacks. This ensures pipeline stability even if the LLM output format slightly deviates.

2. Agent Workflow and Tool Usage Strategy

The agent follows a deterministic four-step workflow, executing a "funnel" strategy that narrows down from broad searches to granular, itemized discrepancy detection.

Step 1: Information Extraction

The agent reads the CV text and extracts key entities into a structured dictionary. This establishes the baseline "claims" made by the candidate.

Step 2: Tool Usage and Profile Searching

The agent utilizes the MCP tools strategically to verify the candidate's digital footprint:

- Facebook (Identity Verification): Uses search_facebook_users with fuzzy=True to handle potential typos in the CV. It retrieves the top candidates and uses get_facebook_profile to verify basic identity and location claims.
- LinkedIn (Professional Verification): Uses search_linkedin_people , appending the location extracted from the CV to narrow the search space. It then uses get_linkedin_profile to retrieve full professional and educational histories.

Step 3: Discrepancy Detection

The agent conducts a cross-examination using specialized prompts:

- Identity Check: Compares CV names and locations against Facebook/LinkedIn data.
- Experience & Education Check: Compares CV claims specifically against LinkedIn data. The LLM identifies discrepancies (e.g., differing job titles, exaggerated tenure, missing companies) and assigns a severity level (high, medium, low).

Step 4: Deterministic Scoring Algorithm

To meet the evaluation requirements (decision threshold of 0.5), the system utilizes a weighted scoring mechanism:

- Base Score: Starts at 0.35 (assuming an unverified CV is problematic by default).
- Verification Bonuses: A matched LinkedIn profile adds up to +0.30, and a Facebook match adds up to +0.20 (weighted by the LLM's match confidence). Maximum baseline score is 0.85.
- Discrepancy Penalties: Points are deducted based on severity: High (-0.20), Medium (-0.10), Low (-0.05).
- Final Decision: The score is clamped between [0.0, 1.0]. A score > 0.5 triggers a "Valid CV" decision, while <= 0.5 flags a "Problematic CV".

Sample Verification Results

The system was evaluated against the 5 provided sample CVs. The evaluation achieved a 100% accuracy rate against the ground truth labels [1, 1, 1, 0, 0]. Below are the summarized verification results:

```
================================================================================
📋 VERIFICATION REPORT: CV_1.pdf
================================================================================

📌 SUMMARY
----------------------------------------
  Candidate Name: John Smith
  Claimed Location: Singapore, Singapore
  Education: 1 entries
  Experience: 1 entries

📘 SOCIAL MEDIA MATCHES
----------------------------------------
  Facebook: ✓ Found
    - Name: John Smith
    - Location: Singapore, Singapore
  LinkedIn: ✓ Found
    - Name: John Smith
    - Headline: Marketing Professional

⚠️ DISCREPANCIES
----------------------------------------
  Total: 0 | High: 0 | Medium: 0 | Low: 0
  ✅ No discrepancies found

📊 SCORE & DECISION
----------------------------------------
  Score: 0.80 / 1.00
  Decision: Valid CV
```

```
================================================================================
📋 VERIFICATION REPORT: CV_2.pdf
================================================================================

📌 SUMMARY
----------------------------------------
  Candidate Name: Minh Pham
  Claimed Location: Beijing, China
  Education: 1 entries
  Experience: 2 entries

📘 SOCIAL MEDIA MATCHES
----------------------------------------
  Facebook: ✓ Found
    - Name: Minh Pham
    - Location: Beijing, China
  LinkedIn: ✓ Found
    - Name: Minh Pham
    - Headline: Design Professional

⚠️ DISCREPANCIES
----------------------------------------
  Total: 0 | High: 0 | Medium: 0 | Low: 0
  ✅ No discrepancies found

📊 SCORE & DECISION
----------------------------------------
  Score: 0.77 / 1.00
  Decision: Valid CV
```

```
================================================================================
📋 VERIFICATION REPORT: CV_3.pdf
================================================================================

📌 SUMMARY
----------------------------------------
  Candidate Name: Wei Zhang
  Claimed Location: Munich, Germany
  Education: 1 entries
  Experience: 1 entries

📘 SOCIAL MEDIA MATCHES
----------------------------------------
  Facebook: ✓ Found
    - Name: Wei Zhang
    - Location: Munich, Germany
  LinkedIn: ✓ Found
    - Name: Wei Zhang
    - Headline: Consulting Professional

⚠️ DISCREPANCIES
----------------------------------------
  Total: 0 | High: 0 | Medium: 0 | Low: 0
  ✅ No discrepancies found

📊 SCORE & DECISION
----------------------------------------
  Score: 0.82 / 1.00
  Decision: Valid CV
```

```
================================================================================
📋 VERIFICATION REPORT: CV_4.pdf
================================================================================

📌 SUMMARY
----------------------------------------
  Candidate Name: Rahul Sharma
  Claimed Location: Singapore
  Education: 1 entries
  Experience: 2 entries

📘 SOCIAL MEDIA MATCHES
----------------------------------------
  Facebook: ✗ Not Found
  LinkedIn: ✗ Not Found

⚠️ DISCREPANCIES
----------------------------------------
  Total: 0 | High: 0 | Medium: 0 | Low: 0
  ✅ No discrepancies found

📊 SCORE & DECISION
----------------------------------------
  Score: 0.35 / 1.00
  Decision: Problematic CV
```

```
================================================================================
📋 VERIFICATION REPORT: CV_5.pdf
================================================================================

📌 SUMMARY
----------------------------------------
  Candidate Name: Rahul Sharma
  Claimed Location: London
  Education: 1 entries
  Experience: 4 entries

📘 SOCIAL MEDIA MATCHES
----------------------------------------
  Facebook: ✓ Found
    - Name: Rahul Sharma
    - Location: London, UK
  LinkedIn: ✗ Not Found

⚠️ DISCREPANCIES
----------------------------------------
  Total: 0 | High: 0 | Medium: 0 | Low: 0
  ✅ No discrepancies found

📊 SCORE & DECISION
----------------------------------------
  Score: 0.43 / 1.00
  Decision: Problematic CV
```